

Bjarne R Bartlett PhD

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Highly adaptable educator and researcher pioneering the integration of generative AI into the SUNY STEM curriculum to democratize access to bioinformatics. Proven success in leveraging Large Language Models (LLMs) as educational intermediaries to shift student paradigms from "learning to code" to "coding to learn," making complex genomic data accessible without traditional programming prerequisites. Focused on cultivating crucial digital literacy, critical thinking, and algorithmic evaluation skills to prepare the next generation of scientific thinkers.

EMPLOYMENT

Farmingdale State College

- Assistant Professor *2026-present*
- PRODiG+ Fellow *2024-2026*

Oregon State University

- Postdoctoral Fellow: Chris Curtin Lab *2022-2024*

Viracta Therapeutics

- Bioinformatics Consultant *2022-2024*

Personal Genome Diagnostics

- Consulting Laboratory Technician *2015-2017*

Johns Hopkins School of Medicine

- Research Specialist *2012-2015*
- Research Technician *2015-2017*

EDUCATION

Gettysburg College

- Bachelor's of Science: Biology *2012*

The University of Hawai'i at Mānoa

- Doctor of Philosophy: Molecular Biosciences and Bioengineering *2022*
- Dissertation: "Data Science for Molecular Genetics and Communication in the Natural Sciences"

TEACHING

Farmingdale State College

- FYE 101: First Year Experience *2026-present*
- BIO 120: General Biology (For Non-Majors) *2024-present*
- BIO 130: Biological Principles I (For Majors) *2024-present*
- BIO 318: Medical Microbiology *2024-present*
- BIO 345: Bioinformatics *2025-present*

University of Hawai'i

- INBRE Program: Bioinformatics
- Department of Microbiology 461L: Immunology

PUBLICATIONS

- Merging Subject-Matter with Computational Expertise: Lessons from a Multi-Year Bioinformatics Practicum for Undergraduate Researchers. Valdez A., **Bartlett B.**, et al. *In Preparation*.
- Thibodeau A, Reardon P, **Bartlett B**, Curtin C. Changes in microbial community structure during adaptation of Kombucha symbiotic culture of bacteria and yeast to fermentation of sweet and acid Whey. *Journal of Dairy Science*. 2025 May 1;108(5):4761-84.
- **Bartlett B**, Zamora CM, Bingham JP, Hubbard AS, Tseng M, Runck B, Kantar M. Impact factor does not predict long-term article impact across 15 journals. *Data and Information Management*. 2024 Dec 1;8(4):100079.
- **Bartlett, B.**, Stitt-Bergh, M., Kantar, M., & Bingham, J. P. (2023). A data science practicum to introduce undergraduate students to bioinformatics for research. *Biochemistry and Molecular Biology Education*, 51(5), 520-528.
- Tamrazi, A., Sundaresan, S., Gulati, A., Tan, F. J., Wadhwa, V., **Bartlett, B. R.**, & Diaz Jr, L. A. (2023). Endovascular image-guided sampling of tumor-draining veins provides an enriched source of oncological biomarkers. *Frontiers in Oncology*, 13, 1166.
- **Bartlett, B.**, Cho, A., Laspisa, D., Gore, M. A., & Kantar, M. B. (2023). Genomic resources for Macadamia tetraphylla and an examination of its historic use as a crop resource in Hawaii. *bioRxiv*, 2023-12.
- Paudel, R., **Bartlett, B.**, Zamora, C.M., Keach, J.E., Gutierrez-Coarite, R., Hawkins, J., Ahmad, A., Motomura-Wages, S., Kirk, E.R., Kantar, M.B. and Lamour, K.H., 2022. Breeding and selection of taro (*Colocasia esculenta*) for improved disease-resistance in Hawai'i. *Plants, People, Planet*.
- **Bartlett B**, Gao Z, Schukking M, Menor M, Khadka VS, Fabbri M, Deng Y. The miRNA Profile of Inflammatory Colorectal Tumors Identify TGF- β as a Companion Target for Checkpoint Blockade Immunotherapy. *Frontiers in Cell and Developmental Biology*. 2021 Oct: 2757.
- Fortin J, **Bartlett B**, Kantar M, Tseng M, Mehrabi Z. "Digital technology helps remove gender bias in academia." *Scientometrics*. 2021 May;126(5):4073-81.
- **Bartlett B**, Fortin J, Kantar M, Tseng M, Mehrabi Z. "How altmetrics could help level the playing field for women in STEM." *Nature Index*. 2021 March.
- **Bartlett B**, Zhu Y, Menor M, Khadka VS, Zhang J, Zheng J, Jiang B, Deng Y. "Development of a RNA-Seq based prognostic signature for colon cancer." *International Journal of Computational Biology and Drug Design*. 2020;13(5-6):488-503.
- Georgiadis A, Durham JN, Keefer LA, **Bartlett BR**, Zielonka M, Murphy D, White JR, Lu S, Verner EL, Ruan F, Riley D, Anders RA, Gedvilaite E, Angiuoli S, Jones S, Velculescu VE, Le DT, Diaz Jr. LA, Sausen M. "Noninvasive detection of microsatellite instability and high tumor mutation burden in cancer patients treated with PD-1 blockade." *Clinical Cancer Research*. 2019 Dec 1;25(23):7024-34.
- Smith KN, Llosa NJ, Cottrell TR, Siegel N, Fan H, Suri P, Chan HY, Guo H, Oke T, Awan AH, Verde F, Danilova L, Anagnostou V, Tam AJ, Lubner BS, **Bartlett BR**, Aulakh LK, Sidhom JW, Zhu Q, Sears CL, Cope L, Sharfman WH, Thompson ED, Riemer J, Marrone KA, Naidoo J, Velculescu VE, Forde PM, Vogelstein B, Kinzler KW, Papadopoulos N, Durham JN, Wang H, Le DT, Justesen S, Taube JM, Diaz Jr. LA, Brahmer JR, Pardoll DM, Anders RA, Housseau F. "Persistent mutant oncogene specific T cells in two patients benefitting from anti-PD-1." *Journal for immunotherapy of cancer*. 2019 Dec;7(1):40.
- Llosa NJ, Lubner B, Tam AJ, Smith KN, Siegel N, Awan AH, Fan H, Oke T, Zhang J, Domingue J, Engle EL, Roberts CA, **Bartlett BR**, Aulakh LK, Thompson ED, Taube JM, Durham JN, Sears CL, Le DT, Diaz Jr. LA, Pardoll DM, Wang H, Anders RA, Housseau F. "Intratumoral Adaptive Immunosuppression and Type 17 Immunity in Mismatch Repair Proficient Colorectal Tumors." *Clinical Cancer Research*. 2019 May 6.
- Mandal R, Samstein RM, Lee KW, Havel JJ, Wang H, Krishna C, Sabio EY, Makarov V, Kuo F, Blecua P, Ramaswamy AT, Durham JN, **Bartlett B**, Ma X, Srivastava R, Middha S, Zehir A, Hechtman JF, Morris L GT, Weinhold N, Riaz N, Le DT, Diaz Jr. LA, Chan TA. "Genetic diversity

of tumors with mismatch repair deficiency influences anti-PD-1 immunotherapy response." *Science*. 2019 May 3;364(6439):485-91.*

- Le DT, Durham JN, Smith KN, Wang H, **Bartlett BR**, Aulakh LK, Lu S, Kemberling H, Wilt C, Lubner BS, Wong F, Azad NF, Rucki AA, Laheru D, Donehower R, Zaheer A, Fisher GA, Crocenzi TS, Lee JJ, Greten TF, Duffy AG, Ciombor KK, Eyring AD, Lam BH, Joe A, Kang SP, Holdhoff M, Danilova L, Cope L, Meyer C, Zhou S, Goldberg RM, Armstrong DK, Bever KM, Fader AN, Taube J, Housseau F, Spetzler D, Xiao N, Pardoll DM, Papadopoulos N, Kinzler KW, Eshleman JR, Vogelstein B, Anders RA, Diaz Jr. LA. "Mismatch repair deficiency predicts response of solid tumors to PD-1 blockade." *Science*. 2017 Jul 28;357(6349):409-13. *
- Parpart-Li S, **Bartlett B**, Popoli M, Adleff V, Tucker L, Steinberg R, Georgiadis A, Phallen J, Brahmer JR, Azad NA, Browner I, Laheru D, Velculescu V, Sausen M, Diaz LA Jr. "The effect of preservative and temperature on the analysis of circulating tumor DNA." *Clinical Cancer Research*. 2016; 23 (10): 2471-2477.*
- Le DT, Uram JN, Wang H, **Bartlett BR**, Kemberling H, Eyring AD, Skora AD, Lubner BS, Azad NS, Laheru D, Biedrzycki B, Donehower RC, Zaheer A, Fisher GA, Crocenzi TS, Lee JJ, Duffy SM, Goldberg RM, de la Chapelle A, Koshiji M, Bhaijee F, Huebner T, Hruban RH, Wood LD, Cuka N, Pardoll DM, Papadopoulos N, Kinzler KW, Zhou S, Cornish TC, Taube JM, Anders RA, Eshleman JR, Vogelstein B, Diaz Jr. LA. "PD-1 blockade in tumors with mismatch-repair deficiency." *New England Journal of Medicine*. 2015; 372(26):2509-2520. *
- Bettinger C, Sausen M, Leary RJ, Kinde I, Wang Y, Agrawal N, **Bartlett BR**, Wang H, Lubner B, Alani RM, Antonarakis ES, Azad NS, Bardelli A, Brem H, Cameron JL, Lee CC, Fecher LA, Gallia GL, Gibbs P, Le DT, Giuntoli RL, Goggins M, Hogarty MD, Holdhoff M, Hong SM, Jiao Y, Juhl HH, Kim JJ, Siravegna G, Laheru DA, Lauricella C, Lim M, Lipson EJ, Marie SKN, Netto GJ, Oliner KS, Olivi A, Olsson L, Riggins GJ, Sartore-Bianchi A, Schmidt K, Shih LM, Oba-Shinjo SM, Siena S, Theodorescu D, Tie J, Harkins TT, Veronese S, Wang TL, Weingart JD, Wolfgang CL, Wood LD, Xing D, Hruban RH, Wu J, Allen PJ, Schmidt CM, Choti MA, Velculescu VE, Kinzler KW, Vogelstein B, Papadopoulos N, Diaz Jr. LA. "Detection of circulating tumor DNA in early- and late-stage human malignancies." *Science Translational Medicine*. 2014; 6(224):224ra24. *

*Cited more than 100 times

^Undergraduate Student Author

OUTREACH AND PRESENTATIONS

- **Bartlett BR**. "Democratizing Bioinformatics: Leveraging AI to Bridge the Gap Between Biological Inquiry and Digital Literacy" (2026). General Education Conference. State University of New York.
- **Bartlett BR**. "Toxins, Teas, and Technology: Building Data Bridges Across Biology" (2025). Eleanor Fapohunda Colloquium Series. Farmingdale State College.
- **Bartlett BR**. "Integrating microplastics as part of drug discover research on conotoxins." (2025). SPARK Program, Brookhaven National Lab.
- **Bartlett BR**. "Impact factor does not predict long-term article impact across 15 journals." Presentation to the Celebration of Scholarship, Farmingdale State College (2025).
- **Bartlett BR**. "Beyond MSI: Immune Peptidomics for Personalized Cancer Immunotherapy." (2024). Yale School of Medicine.
- **Bartlett B**, Fortin J, Kantar M, Tseng M, Mehrabi Z. "How altmetrics could help level the playing field for women in STEM." (2021). *Nature Index*. [Link](#).
- **Bartlett BR**. "Using RNA-Seq to Predict Prognosis for Colon Cancer Patients." International Conference on Intelligent Biology and Medicine (2019). Selected for oral presentation.
- Yu Z, Zhu Y, Ai J, **Bartlett B**, Zhang J, Jiang B, Deng Y. "Development of predictive models to distinguish metals from non-metal toxicants, and individual metal from one another." International Conference on Intelligent Biology and Medicine (2019). Selected for oral presentation.

- **Bartlett BR.** "The miRNA Expression Profile of Inflammatory Tumors Reveals a Unique Immune Cell Profile and Potential Companion Targets for Checkpoint Blockade Immunotherapy." AACR-JCA Joint Conference (2019). Selected for poster.
- **Bartlett B,** Khadka V, Menor M, Deng Y. "The miRNA expression profile of inflammatory tumors reveals a unique immune cell profile and potential companion targets for checkpoint blockade immunotherapy." AACR (2019). Selected for poster.
- Georgiadis A, Durham JN, Keefer L, **Bartlett BR,** Zielonka M, Murphy D, White JR, Lu S, Verner E, Ruan F, Riley D. "Analysis of cell-free plasma DNA to identify tumors with microsatellite instability and exceptionally high tumor mutation burden in patients treated with PD-1 blockade." European Journal Of Cancer (2018).
- **Bartlett BR.** "Developing a solid-tumor screening program checkpoint blockade immunotherapy." The University of Hawai'i (2017). Invited speaker.
- Parpart-Li ST, **Bartlett B,** Popoli M, Adleff V, Brahmer J, Azad N, Bonerigo S, Browner I, Ryan A, Velculescu V, Sausen M. "Optimized plasma collection procedures for liquid biopsy analyses in cancer." AACR (2016).
- Le DT, Uram JN, Wang H, **Bartlett B,** Kemberling H, Eyring A, Azad NS, Laheru D, Donehower RC, Crocenzi TS, Goldberg RM. "Programmed death-1 blockade in mismatch repair deficient colorectal cancer." Journal of Clinical Oncology (ASCO Meeting Abstracts) 2016
- Diaz LA, Uram JN, Wang H, **Bartlett B,** Kemberling H, Eyring A, Azad NS, Dausen T, Laheru D, Lee JJ, Crocenzi TS. "Programmed death-1 blockade in mismatch repair deficient cancer independent of tumor histology." Journal of Clinical Oncology (ASCO Meeting Abstracts) 2016
- Le DT, Uram JN, Wang H, Kemberling H, Eyring A, **Bartlett B,** Goldberg R, Crocenzi TS, Fisher GA, Lee JJ, Greten TF, Laheru DA, Azad NS, Donehower RC, Luber B, Koshiji M, Eshleman JR, Anders RA, Vogelstein B, Diaz LA. "PD-1 blockade in mismatch repair deficient non-colorectal gastrointestinal cancers." Journal of Clinical Oncology (ASCO Meeting Abstracts). 2016;34;(abstracts 195).
- Parpart-Li S, **Bartlett B,** Brahmer J, Azad N, Bonerigo S, Browner I, Ryan A, Velculescu V, Sausen M, Diaz L. "Optimized plasma collection procedures for cell-free DNA analysis in advances malignancies." Circulating nucleic acids in plasma and serum (CNAPS) IX congress (2015). Selected for oral presentation.
- Le DT, Uram JN, Wang H, **Bartlett B,** Kemberling H, Eyring A, Skora A, Azad NS, Laheru DA, Donehower RC, Luber B, Crocenzi TS, Fisher GA, Duffy SM, Lee JJ, Koshiji M, Eshleman JR, Anders RA, Vogelstein B, Diaz LA. "PD-1 blockade in tumors with mismatch repair deficiency." Journal of Clinical Oncology (ASCO Meeting Abstracts). 2015;33(15);(abstracts LBA100).
- Wang JS, Sausen M, Parpart-Li S, Murphy DM, Velculescu VE, Wood LD, Solt-Linville S, Sugar E, **Bartlett B,** Blair C, Dausen T, Jaffee EM, Hruban RH, Laheru D, Diaz LA. "Circulating tumor DNA (ctDNA) as a prognostic marker for recurrence in resected pancreas cancer." Journal of Clinical Oncology (ASCO Meeting Abstracts). 2015;33(15):11025.
- Bettgowda C, Sausen M, Leary R, Kinde I, Agrawal N, **Bartlett B,** Wang H, Luber B, Kinzler K, Vogelstein B, Papadopoulos N, Diaz L. "Abstract 5606: Detection of circulating tumor DNA in early and late stage human malignancies." *Cancer Research*. 2014; 74(19 Supplement):5606.
- **Bartlett BR,** Keane B, Fong P. "The Effect of Venlofaxin on the Burrowing Speed of *Corbicula fluminea*." Gettysburg College capstone poster presentation. 2011. (Advisor Dr. Peter Fong).

HONORS AND AWARDS

Best Papers, Data and Information Management

2024

- "Impact factor does not predict long-term article impact across 15 journals"

CI-TRACS Data Science Fellow, University of Hawai'i Cyberinfrastructure

2021

- Selected to be a fellow with the Hawai'i Data Science Institute as part of a \$1 million National Science Foundation grant to increase student cyberinfrastructure skills in climate science.
- Article: [\\$1M supports resilience to climate change through cyberinfrastructure training](#)

Featured Researcher, University of Hawai'i News 2021
 • Article: [New metrics could reduce gender gap in STEM fields](#)

Merit-Based Scholarship, MBBE 2020
 • Received a \$1,000.00, merit-based scholarship for academic productivity as a graduate student

Recognition for 50 Years of Broadcasting, Honolulu City Council 2019
 • KTUH Honolulu recognized for 50 years of college radio broadcasting

Scholar in Training Award Recipient, AACR-JCA Joint Conference 2019
 • Received an \$800.00, merit-based award from AACR to present Master's Degree thesis project:
 "The miRNA Expression Profile of Inflammatory Tumors Reveals a Unique Immune Cell Profile
 and Potential Companion Targets for Checkpoint Blockade Immunotherapy."

Travel Award Recipient, The International Conference on Intelligent Biology & Medicine 2019
 • Received a \$600.00, merit-based award from ICIBM to present master's research project:
 "Using RNA-Seq to Predict Prognosis for Colon Cancer Patients"

Best College Radio Website, College Broadcasters International 2019
 • As Web Director recognized as a finalist for best college radio website: KTUH.org

SERVICE TO THE DISCIPLINE

- **Editor**, Molecular Techniques, Milne Library Publishing (SUNY Geneseo) 2025
- **Grant Review**, SUNY IITG/OER Impact Grants 2025
- **Judge**, Long Island Science and Engineering Fair 2026
- **Meeting Volunteer**, American Association for the Advancement of Science (Seattle, WA) 2020
- **Meeting Coordinator**, International Conference of Translational Medicine (Honolulu, HI) 2019
- **Meeting Volunteer**, Engineering in Medicine and Biology Conference (Honolulu, HI) 2018
- **Meeting Secretary**, Circulating Nucleic Acids in Plasma and Serum Meeting VIII 2013
- **Member**, The International Society on Toxinology 2024-present
- **Member**, The American Chemical Society 2024-present
- **Member**, Institute of Electrical and Electronics Engineers 2017-present
- **Associate Member**, The American Association for Cancer Research 2016-present
- **Member**, American Association for the Advancement of Science 2016-present

ADVISING AND MENTORSHIP

- **Kevin Velasquez**, Farmingdale State College 2026-present
- **Sofia Zahir**, Farmingdale State College 2026-present
- **Benjamin Re**, Farmingdale State College 2024-2026
- **Angel Valdez**, University of Hawai'i 2021
- **Carter Zamora (Gettysburg College)**, University of Hawai'i, REU 2021
- **Carolyn Zhao & Erica Dawson**, Johns Hopkins School of Medicine 2013-2017
- **Peer Learning Associate**, Gettysburg College 2010-2012

FUNDING RECEIVED

- **Nexus Center**, Farmingdale State College 2025
- **SEED IDEAS Diversity Grant**, Kantar Lab 2020
- **SEED IDEAS Diversity Grant**, KTUH Honolulu 2019

- **AAAS Science Outreach Grant**, Kantar Lab

2018

COMMUNITY SERVICE/LEADERSHIP

- **Student Media Advisory Board, Farmingdale State College** *2025-present*
- **Yonkers Military Band** *2025-present*
- **Lehman College and Community Band** *2025-present*
- **General Manager**, KTUH Honolulu *2018-2021*
- **Volunteer for Speaking Science**, GENE-ius Day Summer Program *2019*
- **Team Captain**, Swim Across America Baltimore *2012-2015*
- **Co-director Planning Committee**, Johns Hopkins Oncology Survivor Clinic *2012-2015*
- **Resident Assistant**, Gettysburg College *2009-2012*

REFERENCES

- Available upon request.